TTest

What is Ttest?

A t-test is an inferential statistical tool used to determine if there is a significant difference between the means (averages) of two groups, which may be related or independent.

Types of T-Test:

- **Paired Ttest**
- **Unpaired Ttest**

Paired Ttest:

A paired t-test determines if the mean difference between two related (paired) measurements is significantly different from zero. It is used to compare data from the same subjects under two conditions, such as "before-and-after" treatment scenarios, "pre-test/post-test" studies, or "matched pairs" to determine if a significant change occurred.

Paired Ttest Example:

Player	Max Vertical Jump Before Training Program	Max Vertical Jump After Training Program	Difference
Player 1	22	24	-2
Player 2	20	22	-2
Player 3	19	19	0
Player 4	24	22	2
Player 5	25	28	-3
Player 6	25	26	-1
Player 7	28	28	0
Player 8	22	24	-2
Player 9	30	30	0
Player 10	27	29	-2
Player 11	24	25	-1
Player 12	18	20	-2
Player 13	16	17	-1
Player 14	19	18	1
Player 15	19	18	1
Player 16	28	28	0
Player 17	24	26	-2
Player 18	25	27	-2
Player 19	25	27	-2
Player 20	23	24	-1
		Mean of differences	**-0.950**
		Std. dev. of differences	**1.317**

Assignment of Paired Ttest:

```
# On male student with same group and with different condition has secured mark of e_test and mba_p mark
from scipy.stats import ttest_rel
male=dataset[dataset['gender']=='M']['etest_p']
male1=dataset[dataset['gender']=='M']['mba_p']
ttest_rel(male,male1)
```

```
TtestResult(statistic=10.558377508518305, pvalue=1.814031908498205e-19, df=138)
```

Here as per the Ttest result for statistic of 10.558377 and the **pvalue** is 1.81403190 which is more than the high probability value which should be within the significant range of 0.5. There is highly significant difference between the etest_p marks and the mba_p marks for male students. So we reject the null hypothesis.

Unpaired Ttest:

An unpaired t-test (or independent samples t-test) is a statistical method used to compare the means of two independent, unrelated groups to determine if there is a statistically significant difference between them. It tests whether the gap between sample averages reflects a real-world difference or random chance.

Examples of Unpaired Ttest:

Studying Technique #1

	Exam Grade
Student #1	77
Student #2	79
Student #3	83
Student #4	84
Student #5	84
Student #6	87
Student #7	89
Student #8	90
Student #9	94
Student #10	95

Studying Technique #2

	Exam Grade
Student #1	79
Student #2	84
Student #3	80
Student #4	83
Student #5	83
Student #6	82
Student #7	80
Student #8	91
Student #9	92
Student #10	87

Assignment of Unpaired Ttest:

```
#Assignment of HSC board difference of Others and Central board Salary difference
from scipy.stats import ttest_ind
dataset=dataset.dropna()
others=dataset[dataset['hsc_b']=='Others']['salary']
central=dataset[dataset['hsc_b']=='Central']['salary']
ttest_ind(others,central)
```

```
TtestResult(statistic=0.30570032095155825, pvalue=0.7601313863865756, df=213.0)
```

Here is the Ttest Result of Statistic – 0.305700 and pvalue is 0.7601 which is significantly higher than the standard significant level which is 0.05. There is no statistically significant difference in the salaries between the ‘Others’ and the ‘Central’ board but the difference in means is likely due to random chances.